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Current Experiences in Economic Incentives Boosting Coordinated Fuel Reduction for Wildfire Risk Mitigation in Catalonia (Spain)

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Abstract

In a context of increasing wildfire risk and highly fragmented forest ownership, this work investigates two relatively recent monetary policy instruments in Catalonia that require grouped applications: a subsidy for fuel reduction, which prioritises collective applications in wildfire-strategic areas, and a climate credit system that promotes territorially coordinated, multifunctional forest management that, i.a., decreases wildfire risk through fuel management. Through in-depth interviews with beneficiaries and consultations with key informants, we analysed whether these measures have triggered adjacent forest management, and how they have interacted with joint action rules to facilitate concerted interventions. The qualitative content analysis indicates that these measures represent a significant step towards landscape-level management and that pre-existing forest owners' associations play a crucial role in capturing the available funds. The eligibility of coordination costs is also appreciated for covering the transaction costs of catalysing landowners. Yet, areas with weaker social capital may become disadvantaged if there is no external support for their organisation. These findings contribute to the emerging field of policy tools for effective landscape-level interventions.

Keywords: economic incentive; joint forest management; fuel management; forest owners associations; adjacency externality; wildfire risk mitigation; private forest owners; voluntary Carbon market; CO_2 market

1. Introduction

Wildfires represent one of the greatest risks to Mediterranean forest ecosystems, with significant impacts on biodiversity, water resources, carbon sequestration, and human safety. These events have intensified in recent decades due to climate change and the abandonment of forest management [1,2]. In this context, fire does not recognise property boundaries and, under conditions of fragmented landscape ownership, collaboration among landowners represents a key strategy for wildfire risk management [3,4]. However, despite its public interest, free-riding behaviour [5] may emerge when private landowners have a dominant strategy of not undertaking fuel reduction.

In Europe, 60% of the forest area is privately owned [6]; of these private owners, 61% own properties of less than one hectare, and only 1% own properties over 50 hectares. In Spain, 72% of forest land is privately owned, and 98.9% of these properties are smaller than 10 hectares [7]. Small parcels, which provide low economic returns from wood,

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together with the decline of rural lifestyles, constitute factors that contribute to landowners' disconnection from their forest properties [8]. As a result, limited effort is invested in forest management, leading to fuel accumulation that increases wildfire risk and related issues such as biodiversity loss and degradation of ecosystem services.

To address cost-related barriers, several economic incentive mechanisms have been developed to promote forest management and reduce fire risk [9]. In Europe, co-financing support for fuel reduction has traditionally been provided through subsidies framed within the Rural Development Programme of the European Agricultural Fund for Rural Development [10,11]. However, these subsidies do not consider collaboration between landowners or spatial coherence, which limits their effectiveness [11]. Small and disconnected fuel interventions have a limited effect on fire spread behaviour. Achieving coordinated fuel management is particularly challenging in individualistic societies, where landowners show a weak culture of cooperation. This would require, *inter alia*, recurrent interactions, trust building, or technical support to build sufficient and lasting social capital among members [12].

Globally, however, few policy mechanisms adopt a spatial scale perspective. In South Africa [13]: 34 (1) and Australia, landowners are legally liable for fire spread onto neighbouring properties. While this normative framework acknowledges adjacency importance, it does not necessarily promote neighbours' fuel cooperation. Anglo-Saxon countries (Canada, USA, Australia) have developed homeowners' cross-boundary cooperation initiatives focused on the Wildland–Urban Interface [14]. Analogous to our forest/massif focus, Portugal is running initiatives towards landscape-level fuel management—through Forest Intervention Zones (ZIFs) [15], Village Condominium or Land consolidation programmes [16]. With the exception of the study of [3], the assessment and motivations of economic incentives for foresters' coordination remain a niche. In Spain, forest and fire risk management policies are the competence of the regional authorities. In this regard, the Catalonian government developed two market-based mechanisms that promote coordination among landowners with geographically close plots: (i) wildfire mitigation subsidies, and (ii) climate credits. This work analyses the design of these measures and qualitatively assesses their contribution to landowner coordination efforts, particularly with a view toward improving forest management and reducing wildfire risk. Our guiding question was: How do current economic incentives for fuel reduction affect the socio-economic factors underpinning coordination among private forest owners? Our working hypothesis is that monetary support reduces the negative economic balance barrier, and this, combined with the intrinsic interest of landowners to keep a lower risk on their properties (joint motivation), helps in overcoming potential negative attitudes towards collective action. We also hypothesise that areas with stronger social capital show a larger capture of these incentives. This work adds to the body of knowledge on private forest owners' coordination [17].

2. Materials and Methods

2.1. Case Studies

In Catalonia (north-eastern Spain), 76% of forest land is privately owned, and over 50% of owners have plots smaller than one hectare [18], with smaller plots when going south. Two economic incentives were recently developed and implemented, namely:

(i) Wildfire prevention subsidies, funded by the Catalan CO_2 tax on vehicles channelled through the Fund for Natural Patrimony in the 2022–2025 calls. This cost-covering grant covers fuel reduction through thinning (since 2023, also including sapling thinning and shoot selection), and forest road repair works. This mechanism was selected due to its prioritisation criteria, which award higher scores to proposals implemented in strategic wildfire prevention areas within a grouped Forest Management Instru-

ment (IOF). This implies prior cooperation among landowners for its development. Requesting landowners' grouping for subsidy eligibility was identified as an effective cooperation trigger in previous calls on post-windstorm and pest outbreak events in Catalonia.

Our interviewees were beneficiaries of the 2022 call, and at the time of the interviews, the corresponding works were being finalised or certified, while the 2023 call had just finalised the selection process. The guidelines were largely the same across years; however, some minor changes were introduced, which do not affect the core of our analysis.

The potential beneficiaries were forest owners, their temporary groupings, or their associations. Eligible actions must cover a minimum area, with budgets ranging from €5000 to €200,000 per proposal. Zoning prioritisation criteria assign higher scores to Management Promotion Areas (AFG, by its Catalan acronym) and Strategic Management Points (PEG in Catalan) than to other features. Regarding execution, landowners must first advance the costs, after which an intermediate certification allows for an interim payment, followed by a final verification that enables the last payment. The total budget for these subsidies varied slightly from year to year, ranging from approximately 4.9 to almost 4.2 million €.

(ii) Forest Climate Credits, which operate in the voluntary market for carbon and other ecosystem services. The Catalan Government Agreement 270/2023 establishes the Climate Credit system. This system creates a market for climate credits based on multifunctional forest management [19]. Unlike traditional approaches focused solely on reforestation, this system also includes active forestry. Through this scheme, companies and individuals can fully or partially finance projects that improve biodiversity, increase water availability, enhance carbon sequestration, and prevent wildfires.

The management initiatives are articulated through the Forest Projects for Climate Change Mitigation and Adaptation (PROMACC). The PROMACC is an operational, landscape-scale management plan to be implemented within three years, and covers forest properties within the same massif or valley, meaning that landowners participate as a group. Each climate credit is equivalent to one hectare under management, and its price is calculated as the average implementation cost of all the hectares included in the PROMACC. The first PROMACCs were drafted in 2022. Through the previous LIFE CLIMARK project, a specific methodology was developed to estimate the expected benefits in terms of improved biodiversity, optimisation of water consumption in forests, increased CO_2 sequestration in the soil, and reduction in forest fire risk in the managed area [20].

2.2. Conceptual Framework

Wildfire represents for landowners a "common-pool bad", insofar as they cannot be excluded from its spread, yet if one "consumes" it (i.e., extinguishes the fire), then the neighbour does not suffer the bad. Ostrom [21] developed a set of working rules for joint action, which were adapted by [12] for joint forest management, identifying concrete challenges for private forestry coordination (Figure 1). We adopt this framework to analyse how the relevant actors (private forest owners) handle the working rules leading to joint fuel management bolstered by the subsidy or the credit incentives. This entails setting up boundary rules (on location or duration of the joint commitment), position rules (on representativeness and efficiency of the catalyser), and authority rules on agreed actions (on the formal interactions, such as planning mechanisms or wood/credit trading). These actions are mediated by the information rules (transparency or communication channels), control rules (decision-making procedures or trust to control potential free riders), the payoff rules incentivising the action (through perceived fairness and efficiency among the group), and the scope rules in alignment with the expected outcomes of the intervention (as joint motivation and long-term vision).

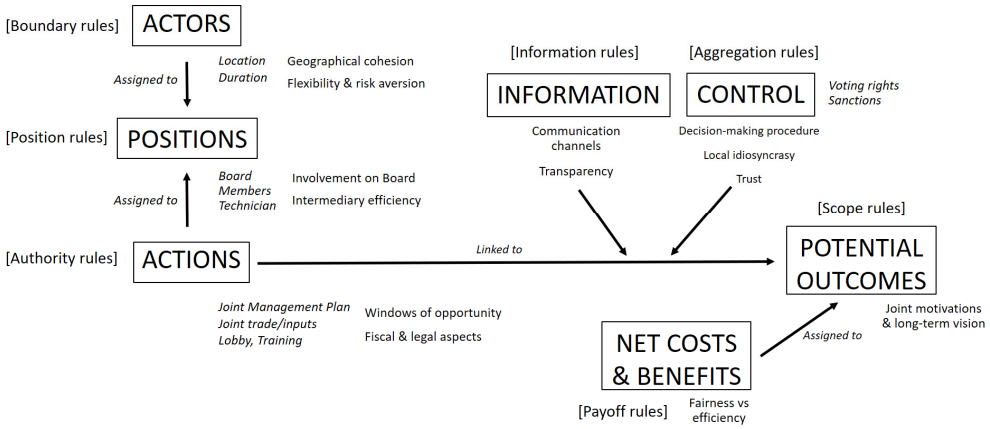


Figure 1. Conceptual framework for joint action, with its elements (boxes), their related working rules in [], the identified joint forest management challenges (normal), and key related variables (italics). Adapted from [21].

2.3. Methodological Approach and Analysis

Our research is based on a qualitative approach, focused on gaining an in-depth understanding of the social and organisational processes underlying the participation of forest owners in two economic incentive mechanisms in Catalonia (our case studies). For this purpose, semi-structured interviews were chosen, as this method allows the capture of not only factual information, but also participants' perceptions, motivations, and subjective experiences [22]. This technique proved especially suitable for exploring aspects such as how the owners got in touch, decision-making within the associations, management of the payment for actions, the process of receiving the subsidies or the credit revenues, as well as the barriers encountered in the process.

Prior to conducting the interviews, a detailed review of the technical and legal documents available on both incentive mechanisms was carried out. This review made it possible (i) to understand the institutional and operational design of the subsidies and credits, (ii) to identify key variables to be explored in the interviews, and consequently (iii) to formulate the interview guide (Supplementary Materials).

A total of nine in-depth interviews were conducted with eleven private forest owners and one association technician between the end of 2023 and early 2024, distributed across the four provinces of Catalonia. This distribution made it possible to capture the diversity of situations resulting from different private forest ownership structures and geographic characteristics across the territory. Interviewees were selected purposively, focusing on individuals who had been major beneficiaries of the 2022 subsidies or who had participated in the LIFE Climark project (five associations at that time). Our sample represented 41% of successful grouped subsidy applications (N = 21) and 80% of active participants in the climate credit market by that time, most of whom had also taken part in the public subsidies. This allowed for an integrated view of both mechanisms and ensured relevance and representativeness in relation to the study's scope. The number of interviews was determined by theoretical saturation [23]; once this sample was gathered, no new relevant information emerged with respect to the study's objectives. Interviewees provided written consent to participate in the interviews and to be recorded. Additional consultations were conducted with one representative of the Catalan Fire Service and two forest administration (Forest Ownership Centre, CPF) technicians in charge of the mechanisms-who provided in 2025 the map data of executed interventions.

For data analysis, the interviews were fully transcribed and subsequently analysed using thematic coding. The MAXQDA 2022 software was used, which allowed for the systematic coding of text fragments. The codes emerged from the conceptual framework

(deductive approach), as well as from the content collected in the interviews themselves (inductive approach), which allowed for a contextualised interpretation of the information. Coding was carried out by the same researcher who conducted the interviews, with continuous comparison with a second researcher to ensure inter-coder reliability. For the presentation of the results, the names, associations, and locations of the participants were anonymised using the code Forest Owners Association (FOA), followed by an identification number.

3. Results

3.1. Effects of Incentives on Spatial Coordination

Overall, our findings suggest that both mechanisms have fostered formal cooperation among forest owners, specifically through the joint implementation of silvicultural works, but to a lesser extent, informal cooperation, as in most cases—with only a few exceptions—those who participated were already part of existing associations and it was through that affiliation that they accessed the incentives. As a result, the incentives have helped strengthen collaboration among forest owners by creating beneficial opportunities within their networks. These opportunities may also encourage other landowners to connect with their neighbours and engage in similar initiatives in the future.

3.1.1. Subsidies for Fuel Management

- Effectiveness

The total strategic area for wildfire risk mitigation spans over 151,195.99 ha. Thus, overall, the managed area after three calls covers 3.87% of the designated strategic areas (Table 1).

Table 1. Number of beneficiaries, budget designated and forest area managed for the 2022 to 2024 calls of the fuel reduction subsidies. Source: CPF.

Subsidy Calls	2022	2023	2024	Accumulated
Available budget (€)	4,497,534.51	4,197,534.51	4,929,675.43	13,624,744.45
Amount granted (€)	4,497,499.04	4,465,796.94	4,923,331.30	8,963,295.98
Selected applications	54	51	Not available	105
Fire prevention area managed (ha)	1339.77	1237.91	3275.58	5853.26

- Promotion of Spatial Coordination

Although the grant guidelines require "*spatial continuity within strategic areas*" our interviews revealed that this does not always translate into adjacent interventions. While this flexibility may have helped build applications with nearby, accessible, and motivated landowners, it may not be entirely effective, given that gaps between treated parcels can allow fire to spread. The map analysis indicates that the average minimum distance between plots within subsidised groups is 287 m. Figure 2 shows the subsidised parcels for fuel reduction (shown in orange) within the strategic areas for fire prevention (Priority Protection Perimeters, shown in green). As can be observed, only in some cases there is adjacency between the intervened parcels. On a positive note, for the 2024 grant round, maximum points were awarded to applications in areas where work had already been carried out in 2022 and 2023. This could help landowners complete previously untreated parcels and spatially concentrate financial support, reinforcing the "oil spot effect".

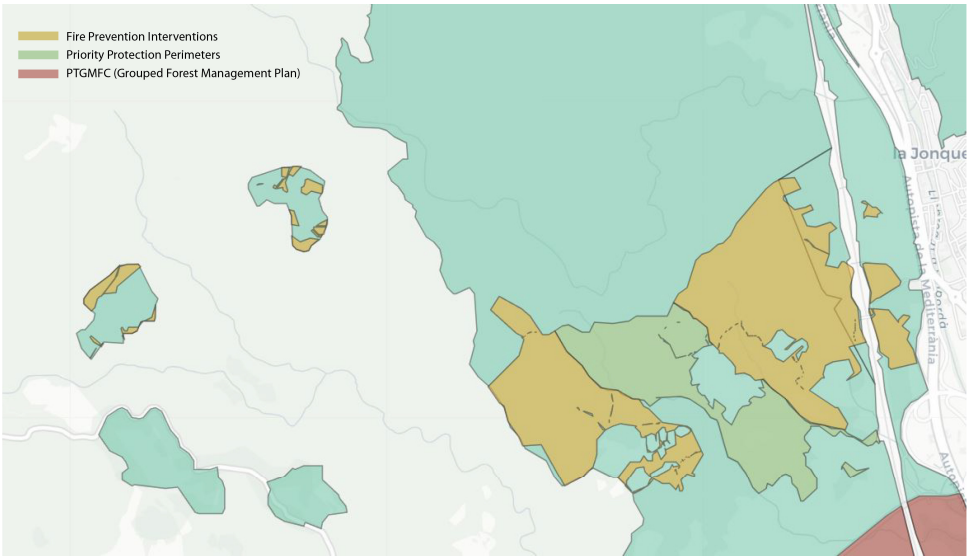


Figure 2. Example of strategic wildfire prevention areas (green) and subsidised plots (orange) carried out. Source: own elaboration based on 2025 CPF data.

- Functioning of the grant

Our interviewees were beneficiaries of the 2022 grants. The results show that this line of support was well received by landowners, firefighters and the administration. Whereas the first call saw a limited number of applications—allowing most eligible applicants to receive funding, the second call in 2023 saw a threefold increase in demand relative to the available budget, making the prioritisation criteria crucial in selecting beneficiaries. The funded actions were generally aligned with what landowners view as necessary for reducing fire risk: thinning and shrub layer reduction, as well as forest road repairs. However, we noted two views regarding the desired level of biomass reduction: while some prefer a more intense thinning, the technical criteria stipulated in the ORGEST guidelines [24] require the maintenance of a certain canopy cover to reduce the resprouting of shrubs in the medium term. Overall, participants expressed satisfaction with how the programme worked and with the outcomes.

That said, we identified areas for improvement in both effectiveness and fairness. Some fairness concerns arose from cases where landowners were well-coordinated and motivated to participate but were left out due to specific constraints.

One such issue stems from prioritisation criteria that favour proposals covering a larger percentage of strategic zones, while simultaneously imposing a €200,000 cap on funding. Since the delineation of strategic zones is externally defined, landowners in larger zones are at a disadvantage: they face higher costs to cover the full area, meaning they end up covering a smaller percentage with the same budget, reducing their chances of selection.

> *"We're talking about work that costs 3000 euros per hectare, right? If I have very large areas and at most, I can request 200,000 euros, if the unit cost is very high, even if I agreed with all the landowners, I still can't act on everything, because it's not enough, right?"* —FOA9

Other potentially exclusionary criteria include the requirement that all parcels involved must have an approved Forest Management Instrument (IOF) or be part of a grouped IOF. This requires prior investment in building the area's social capital. Without the support of an external organisation, this is particularly difficult in regions with highly fragmented land ownership—such as the southernmost province of Tarragona. Further exacerbating this situation, higher scores are also assigned to parcels with sustainable forest management

certification, something that small landowners often lack due to the low revenues from their forests. As a result, this incentive has been awarded less frequently in areas with small-scale ownership, limited profitability or no prior organisational structure. The criteria for 2023 and 2024 were slightly adjusted to accept applications from parcels that were in the process of joining a grouped IOF, even if they did not yet have individual plans.

> *"Because in X I've already told you that we are small landowners. Of course, how do you expect this to have forest certification? They don't have it, nor do they have a technical plan, nor have they downloaded the joint plan. . . This is nonsense, this is like saying in X we give you nothing"* —FOA3

A widespread concern—also noted in other funding programmes—is the payment method. Beneficiaries must advance the costs, which are reimbursed based on the certification of completed work. While partial justifications (up to 25%) are allowed, this upfront cost requirement excludes those who lack the capital to advance the funds. For many, applying through an association is, for this reason, more viable, as associations may have better cash flow to handle these costs.

> *"So, at the Generalitat, in order to get paid you must have paid first. And paying means that you have to take the money from one place or another"* —FOA1

From a tax perspective, applying through an association involves some advantages and disadvantages: From the viewpoint of the associations, unless they conduct some economic activity, they need to request a specific tax exception; otherwise, they need to bear 21% VAT on the received funding, which is not an eligible cost. From the viewpoint of the landowner, applying as an association instead of as a group avoids potential tax auditing by the Treasury. This is because when participating as a group, the person acting as group representative receives the total sum, and even if s/he must transfer the corresponding part to the other beneficiaries, this sudden income triggers a warning on the Treasury to conduct an inspection. While this inspection can be easily resolved, it implies some related transaction costs for that person.

> *"Because, as it is not a legally constituted entity, they do not have a current account and a tax identification number for all of them, but rather each one is an individual, although there is one who puts it in his name and gives his bank account. And this, apart from the inconvenience, creates a certain fiscal insecurity. Because the owner who says, I will deposit here the 200,000 euros from the income of all four of them, their advisers tell them, oh, be careful with this deposit, because it is a lot of money. Yes, when it comes to paying taxes, they will charge him in full"* —FOA9

However, associations are only allowed to submit one application. Therefore, some associations chose to apply for some parcels through the association and encouraged other individuals to submit applications for other parcels as informal groups of individuals.

The aid covers 100% of the intervention costs within set ranges. Another eligible cost is project management up to 6%, a cost which can be used to cover coordination costs related to application preparation and technical supervision—an aspect that is widely appreciated compared with other forestry subsidies.

3.1.2. Forest Climate Credits

- Effectiveness

As of January 2026, a total of ten PROMACCs have been approved and are able to generate 542 climate credits—that is, equivalent hectares that can be managed once the corresponding credits are sold. To date, 293 credits have been sold, enabling the full implementation of four of the PROMACCs. The total fund mobilised through the

purchase of climate credits amounts to 1,088,071 €. Seven (27%) out of the twenty-six purchasing entities have acquired credits in more than one year, while the remainder made one-off purchases.

Table 2 shows the locations of the PROMACCs, the number of credits on sale and already traded, the year of sale, the credit price and the amount of money mobilised in each area.

Table 2. Climate credits results by December 2025.

PROMACC Name	Climate Credits		Credits Sold				Amount Mobilised (€)
	For Sale	Acquired	2022	2023	2024	2025	
Serra de Collserola	40	40	9	31	0	0	137,800
AFG Les Arenes	45	45	15	30	0	0	125,640
Vall de Lord	108	108	0	0	1	107	443,124
Conca Segre-Rialb	80	12	0	0	9	3	38,472
Serres Miralles-Ancosa	34	6	0	6	0	0	20,922
Vall de la Muga	58	17	0	7	9	1	75,293
Les Gavarres	35	0	0	0	0	0	0
Vall de Bianya	30	30	0	0	30	0	98,130
Collserola V. Occidental 1	56	35	0	0	0	27	142,030
Collserola Baix Llobregat 1	56	0	0	0	0	0	0
Total	542	293	24	74	49	138	1,088,071

- Promotion of spatial coordination

There is no adjacency criterion for the actions to be carried out, nor is there any requirement to be implemented in strategic areas for wildfire prevention. However, as shown in Figure 3, many actions are planned in strategic wildfire-prevention areas (shown in red, orange, and green), and adjacency can be observed in some of the treated stands. The map analysis of PROMACCs shows an average distance among plots of 2192 m, with great variation across PROMACCs [0–10,285]. A key strength of this mechanism is that there is an underlying reflection at the landscape level on where to prioritise actions in order to maximise the ecosystem services provided. In most areas, wildfire risk reduction prompts landowners' engagement—even if this objective is only "indirectly sold" afterwards as part of the credit.

> *"FOA1: Yes, the other thing about Climark is that it is not the same management, although it also favours fire prevention, obviously, it is not such a drastic action, like. . .*
>
> *I: It is not so focused.*
>
> *FOA1: On fire prevention, the magnitude of the clearings and so on, but rather more focused on restructuring a forest with its model more. . . better than it should have. And furthermore, the big difference is that all the grants that we have today, whether from the Provincial Council or from the Generalitat, in recent years as well, the department has launched important lines of fire prevention, and moreover very much designed for the associations, all these grants are restricted to the priority areas. Outside the priority areas, practically, you can do very little, and one of Climark's objectives is that we can act with different actions, isn't it?"*

- Functioning of the incentive

To apply, landowners need to establish a collaboration agreement with the PROMACC promoter entity. They should also agree on the silvicultural work to be carried out on their land and formally authorise them. They need to collaborate to reach a minimum of 50 hectares planned at the landscape level in order to meet the PROMACC minimum-area requirement. However, the actual works linked to the credit can take place over a smaller area (for example, Vall de Bianya PROMACC only works on 30 hectares). Projects larger

than 100 hectares of intervention are not recommended because of the difficulty of securing sufficient funding for all the planned credits.

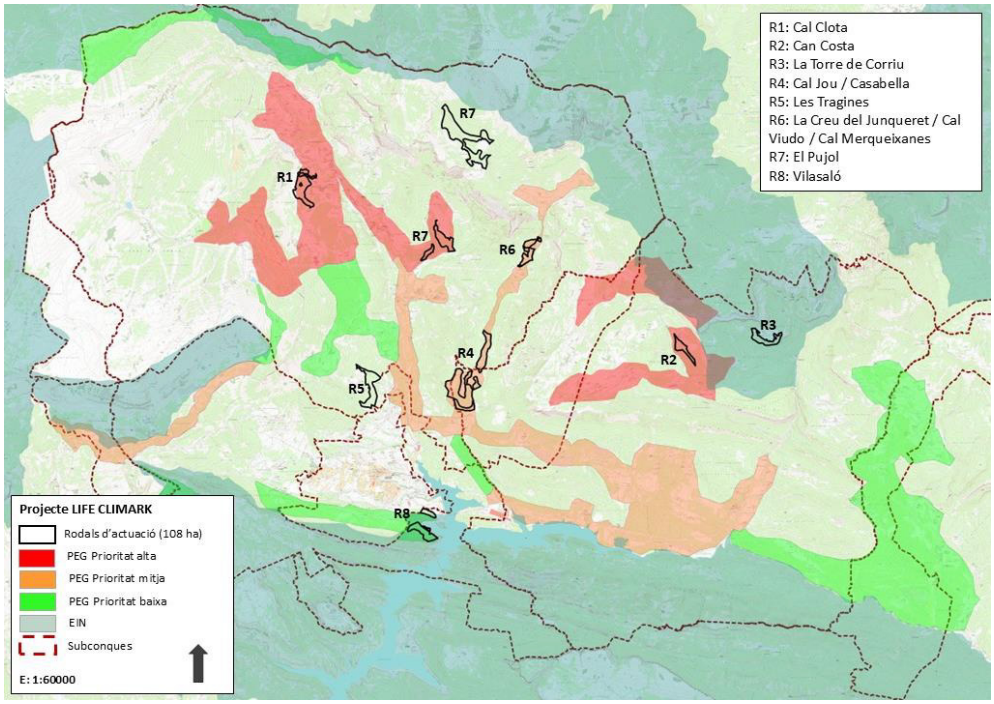


Figure 3. Vall de Lord PROMACC, showing the plots generating credits (R), and the layer of wildfire strategic zones: from higher (red) to lower (green) priority. Source: extracted from https://creditsclimaticsforestals.cat/ (accessed on 4 May 2025).

One of the strengths of this model is its flexibility, as it can be tailored to the specific needs and concerns of each territory. Our observations show that the co-designed nature of PROMACC enables participation by a wide range of local actors, helping to identify and prioritise the most relevant goals for each community.

Interviewees expressed a very positive response towards climate credits, as they offer an alternative for managing their forests without relying on public subsidies and involve new private actors in forestry funding. Some emphasised that this mechanism makes them proud of the work undertaken, as it is perceived as a meaningful contribution to a responsibility shared with society.

> *"We believe that the ones who should pay for forest work are the members of society. And within society, who has the money? Companies. We believe that it should be the companies that provide the funding for this management."* —FOA7

> *"If you find profitability, then the forest takes care of itself. Because if you have to rely on subsidies, it's not sustainable; it's bread for today but hunger for tomorrow."* —FOA3

This system, described as "very well thought out" by some respondents, focuses on multifunctional management and, although it does not focus exclusively on reducing fire risk, has proven to be a key catalyst of formal cooperation between private landowners and other territorial actors such as municipalities, Forest Defence Groups, or local businesses. There is considerable expectation associated with this model, which aims to encourage companies that already buy emission rights or undertake forestry actions elsewhere to also invest in local forest management. Companies such as Mango and Barcelona's trade fair institution have already participated in financing some credits. The Climate Credit covers the costs of the intervention up to roadside piling. Thus, it is left to the discretion of each PROMACC to decide whether to sell that timber jointly or rather leave the wood there for

each forest owner to be handled individually. Joint timber sales were the most frequently reported approach, which either covers transaction costs or constitutes the only net income for the forest owner.

3.2. On the Ability of the Mechanisms to Encourage Cooperation Among Forest Owners

Both incentives have been found to affect four of the joint-action rules (Table 3), namely the scope rules (in terms of the expected outcome of reducing wildfire risk) and the payoff rules (by balancing the costs and benefits of fuel reduction interventions). In addition, through their design requirements, both instruments enhance the aggregation as well as the boundary rules. The instruments presuppose a spontaneous arrangement of the position and authority rules, whereas information rules are not found to play any critical role. These findings are developed next.

Table 3. Identified interactions between the analysed incentives and joint-action rules for wildfire-risk mitigation.

Joint Action Rules	Interactions Identified with the Analysed Incentives
Aggregation rules	- Incentives facilitated moving from informal to formal collaboration and related inter-personal trust - FOA provides benefits to members through both incentives - Possibly, these incentives trigger more FOA establishment or adhesion in the future
Payoff rules	- Incentives as a zero-cost opportunity for active management - Coordination costs (partially) covered
Scoping rules	- Risk reduction interest triggering joint efforts
Boundary rules	- Plot proximity across fuel management increased but not adjacency ensured
Position and Information rules	- No direct effects identified
Authority rules	- Incentives as windows of opportunity to address shared concerns through available funding - Incentives as reward for previous efforts (pre-existence of FOA, approved joint management plans)

3.2.1. Scope Rules: Effects on Joint Motivation and Attractive Long-Term Vision

Both analysed incentives address forest abandonment, which is commonly perceived as a negative feature among landowners, as they generally value maintaining certain forest structure and reducing associated risks, chiefly wildfires. Risk reduction and the maintenance of a forest structure without an abandoned appearance were reported as the most strongly motivating factors. Both incentives were also designed to enhance non-monetary benefits of cooperation, such as biodiversity improvement, blue water provision, and carbon sequestration. In addition, the grouped requirement encourages forest owners to recognise the relevance of their actions for neighbouring properties, thereby becoming part of a larger system beyond their individual parcels. By making cooperation attractive, these incentives can generate a greater positive impact on the forest landscape as a whole.

3.2.2. Payoff Rules: Effects on Cost–Benefit Balance, and the Fairness vs. Efficiency Challenge

Risk mitigation through fuel management is perceived as a costly intervention, which contributes to its limited implementation. The analysed economic incentives directly address this challenge by transforming a loss into an opportunity for concerted action among landowners. Nevertheless, substantial efforts (i.e., personal and/or monetary costs) to coordinate forest owners are still required to catalyse applications and implementation.

Beyond these coordination costs, interviewees did not refer to other direct or indirect costs, either fixed or variable expenditures, derived from the implementation, verification

and permanence phases. Regarding the preparation of the required joint forest management plan (IOF) for the subsidies, these costs are eligible expenditures under a complementary subsidy line. The cost of preparing the PROMACC plan, however, is not included in the credit pricing structure, even though this could be indirectly embedded (at least to a certain level) through the "project management" item. In both instruments, the procedures for fundraising (subsidy application or finding credit buyers) are not covered, and thus represent an investment by the FOA or the PROMACC promoter entity.

Silvicultural interventions were zoned; therefore, not all forest owners are eligible. In each PROMACC, the intervention areas are, in theory, delimited by the participating landowners. However, interviewees reported reliance on locations suggested by the FOA technician, often to maximise the calculation of water, CO_2 or biodiversity gains -and consequently, efficiency. In the subsidy mechanism, the zones were previously delineated by, or in coordination with, the Fire Service. Decisions regarding which plots to prioritise within the eligible zones for subsidy application were also mainly taken by the FOA technician. Given that both instruments are compatible, the same FOA could strategically decide whether a given plot is better suited to the credit mechanism or to subsidies. No conflicts were reported regarding disagreements over which members benefited from these opportunities. Interviewees attributed this lack of conflicts to a general sentiment among landowners that any intervention in their or nearby forests is preferable to inaction. However, tensions could potentially emerge in the future if some landowners feel that, despite being eligible association members, their property is never selected by the board or technician for intervention.

3.2.3. Aggregation Rules: Effects on Trust, Local Idiosyncrasy and Decision-Making Procedure

The two studied mechanisms bolstered trust in FOAs as they have demonstrated their ability to deliver tangible benefits to members (e.g., funding). This effect is particularly relevant in areas where low timber yields render forestry operations largely unprofitable. Some interviewees noted that, at the outset, certain landowners were hesitant to join associations. While this reluctance was sometimes rooted in mistrust of the board or concerns about unequal benefit distribution, it more often reflected a general perception that associations offered little value. Many viewed them as burdensome structures involving fees and time-consuming meetings—a perception counteracted through these incentives.

> *"And then, as a consequence of these first jobs that they started doing, from the protection perimeter grants, and when we could say that we were of some use, that we weren't useless, well, there were people who became interested and five or six more people joined."* —FOA4

The interviewees indicated improved interpersonal trust as a result of the analysed incentives, as they prompted the catalysation of previous social ties into formal collaboration agreements.

Local idiosyncrasies play an important role in the bottom-up design of PROMACC measures, which are therefore tailored to participating forest owners' priorities. The prevailing attitude among interviewees was a preference for some form of intervention in their forest—even when the purpose of the actions was not fully understood—rather than no intervention at all. One interviewee questioned certain technical criteria applied in wildfire-prevention subsidies, suggesting that the fuel-reduction measures implemented were insufficient to meet the stated objectives. This perceived mismatch stemmed either from the application of standardised criteria (ORGEST) that were perceived not to align with local conditions, or from differing perspectives between the administration managing the subsidies and forest owners' understanding of silviculture.

> *"But what I see most is that they [the Administration] haven't provided clear directives. I think that because they rushed, they didn't get in contact either with the firefighters or with anyone else. And the technical way of doing it, they still don't know now. And*

> *we find ourselves, in our view, doing fire prevention actions in Strategic Management Points, and elsewhere, damn it, we carry out actions that are too conservative and don't serve any purpose, but well, we have been doing it."* —FOA3

Regarding climate credits, several interviewees reported not fully understanding the management practices applied. For example, some mentioned being asked to leave deadwood on the forest floor to enhance biodiversity, which contrasts with traditional forest-management practices they had learned. In some cases, landowners also lacked familiarity with the technical terminology used to define these interventions; participation in the PROMACC coordination process allowed them to acquire new knowledge.

> *"FOA7: No, the first day they explained CLIMARK [project] to me, I didn't understand it.*
>
> *I: But apart from not understanding, the first time they told you "dead wood," what did you say? How did you react?*
>
> *FOA7: Well, maybe not, maybe they didn't tell us about the dead wood the first day. Of course, we have been assuming knowledge. When they told me I wasn't surprised. We understood it, yes, and it's not something that, yes, people see as good. And to leave… well, in this forest we did it, we left wood on the ground, we left standing dead wood, we cut trees at 1 metre, yes."*

Most interviewees, and the association members they described, did not expect to obtain economic benefit from their forests, nor were they willing to assume management costs. Forest ownership was not their primary livelihood, and many were largely detached from forest-related activities. For this profile of relatively passive landowners, decision-making within associations was not perceived as a major challenge. Once engaged, landowners entrusted board members with implementing proactive proposals. The main complexity in decision-making processes pertained to reaching the minimum number of participants and obtaining the delegated permission to implement actions.

With regard to wildfire-prevention subsidies, association technicians were generally responsible for deciding which plots would be targeted and what actions would be implemented. Although these decisions were discussed with affected landowners, objections were rare, and issues were seldom raised at general assemblies due to limited interest beyond governing board members.

> *"FOA6: The (…) technical staff of the Provincial Council [guided the decision on which states to reduce fuel].*
>
> *I: So, you didn't really decide much yourselves, right?*
>
> *FOA6: No, because it doesn't make sense for each person to decide; there must be someone superior [sensu knowledgeable] who decides."*

In the case of climate credit participation and PROMACC drafting, processes were reported to be somewhat more participatory in certain instances. Meetings were held to inform members about the project and to identify forest management to prioritise and their consequent ecosystem services gains (e.g., biodiversity enhancement, carbon sequestration). Nevertheless, the actual drafting of the PROMACC and its most operational decisions ultimately fell under the responsibility of the association's technician.

3.2.4. Boundary Rules: Effects on Geographic Cohesion, Actors' Flexibility and Risk Aversion

The design of both the climate credit and the fuel management subsidy set clear geographical boundaries to the target areas: a minimum scale meaningful at the landscape level, zoning, and prioritisation criteria. FOAs are usually organised around a shared territory (such as mountain ranges, valleys, or forested areas). However, it is often difficult for the plots owned by association members to form a continuous, compact forest area

across the landscape. Both incentives studied can play a motivating role in overcoming the geographical cohesion challenge and in encouraging efforts to identify and recruit new members. Subsidies, for example, can encourage neighbouring landowners to reach out to a third party in order to increase their collective score under prioritisation criteria. In the case of carbon credits, landscape-scale planning makes the participation of multiple landowners a necessary condition.

> *"Yes, well, four of us founded it and we made a call. We started working seeing if anyone wanted to join, we organised a meeting with many forest owners, because at the end of the day we would go and carry out the work at these people's homes. No, we didn't go looking for them because they were owners, but because our action had to take place at their property. And the response was mostly to join the association, and now we represent 8000 hectares out of 12,000."* —FOA7

The analysed incentives set time boundaries to which applicants need to adhere. The credit system, based on the voluntary nature of joining a PROMACC, provides flexibility to participate or not, depending on each individual's subjective risk aversion. In the case of PROMACC, these are management plans in which participants commit to implementing the agreed interventions within a maximum of three years and maintaining them for at least fifteen years. For subsidies, beneficiaries are required to refrain from changing the land use of the managed area for a minimum of ten years. Beyond these obligations imposed by the funding entities, whether carbon credit buyers or the Catalan government, participation in these incentives does not restrict the landowner's future decisions regarding forest management, nor does it affect their willingness to cooperate with others in the future.

3.2.5. Position Rules: Effects on Involvement in the Board, and Efficiency of the Intermediary

The weak willingness of members to join FOA executive boards was identified as a common challenge among the interviewed associations. Yet, applying to the incentives as an association or grouping requires efforts either from the board, the technician (if any), or both. They held power in prioritising which parcels were included. The analysed incentives did not address this challenge.

> *"We are 9 people. You can well say that in these associations it's very hard to get people to come to the board. Those who come to the board—some are on the board and hardly ever come, others do come but only come to the board meetings—and from those who come to the board and work from one meeting to the next, you're left with the fingers of one hand cut in half. That's true of any association you talk about."* —FOA1

3.2.6. Authority Rules: Effects on Windows of Opportunity and Fiscal Aspects

Our findings show that many FOAs have emerged as responses to disturbances affecting members' forest land. These disturbances were caused by climatic events, such as windstorms or wildfires, but also by governance-related factors, such as political initiatives to establish a natural park or to regulate hunting activities. The analysed incentives represent an opportunity to tackle shared concerns through available funding, which is also cost-covering, with options to generate a modest economic margin. The applications for the 2023–2025 calls exceeded the available funds, thus showing the high level of interest generated.

FOAs have proven to be an effective mechanism for coordinating landowners and thus participating in the subsidy calls. The design limits the associations' capacity to apply to a maximum of once per call. As a result, some members of the associations formed informal collectives to apply as grouped individuals, but still under the technical and social umbrella of the association. However, larger fiscal and pragmatic benefits for the members are achieved when the FOA remains the legal beneficiary of the grants. Only one climate credit

project was coordinated, not by a FOA but by a foundation dedicated to the conservation of natural heritage. This foundation acted as the PROMACC promoter, bringing together the various stakeholders involved in drafting the plan and taking responsibility for identifying buyers for the credits generated.

3.2.7. Information Rules: Effects on Communication Channels

None of the analysed incentives directly or indirectly affect the information rules. Inter-member communication is typically held through the annual general assembly, or informally among those living or frequently visiting the forest-located municipality. Participation in the subsidy or PROMACC can be operationalised through the signing of authorisations to operate in their holdings, therefore resulting in heterogeneous levels of interest in communication beyond that.

On the other hand, there is frequent communication between the technician (s) and the board members, as well as among the board members themselves. This core group communication takes place by phone or through formal and informal meetings, as needed. These interactions are crucial for the effective functioning of opportunities such as the analysed incentives, including application, implementation, and justification.

4. Discussion and Conclusions

The analysed incentives represent a significant step towards landscape-level forest management, which is essential for the provision of certain ecosystem services, particularly wildfire risk regulation. The focus is shifting from the individual forest owner to a scale that is more effective for addressing such processes. Moving from many kilometres between parcels in traditional forestry subsidies to 287 m between plots in the wildfire prevention subsidy represents a relevant spatial compaction of fuel management -less proximal in the climate credit. Nevertheless, there remains scope to improve effectiveness, equity, and motivational capacity.

The results of this study suggest that, although the grouping requirements embedded in the analysed instruments address the problem of scale mismatches in wildfire risk management [25] by progressing in coordinated fuel management, these instruments still require design refinement to maximise their effectiveness. Fire prevention subsidies do encourage formal coordination efforts among private forest owners; however, their impact is diminished when interventions are not implemented in adjacent areas. This finding is consistent with [26], who identified a discrepancy in Mediterranean Europe between subsidised wildfire prevention activities and the actual dynamics of wildfire behaviour. Therefore, it is essential to design policies and incentives that prioritise interventions in continuous [3], strategically selected areas [10,27], thereby promoting coordinated landscape-scale management. Nevertheless, the optimal landscape-scale of fuel management would consider the agroforest mosaic patterns that determine fire spread [28]. To the best of our knowledge, only the Portuguese Landscape Design and Management Programme [29], the Mosaico project in Extremadura [30], Gran Canaria Mosaico in the Canary Islands, or the Catalan Resilient Landscapes are initiatives covering a multisectoral landscape diagnosis and operational support to its required land management. All these initiatives require a facilitator to coordinate the local-level implementation actors. Yet, they differ in the related economic support, and they do not include adjacency requirements.

Our findings confirm the hypothesis that the analysed economic incentives influence payoff [5] and scope rules, thereby shaping the motivation of private forest owners to act in a coordinated manner for fuel management. However, the incentive design could more precisely define the boundary rules in geographical terms to enhance effectiveness. Moreover, the prioritisation of groupings tends to take for granted the internal configuration

of aggregation rules. Aggregation rules, in turn, depend on position and authority rules, which are strongly influenced by the social fabric of the forested territory.

The current design of the instruments rewards those areas with structured forest ownership, that is, with pre-existing social capital channelled through local networks and FOA. These have been the first movers and have captured the available funds. Since 2000, Barcelona province, with the strongest budget, has devoted resources (technicians, funds) to the establishment of associations and to promoting active forestry through them [31]. In Girona province, landowners have a stronger tradition of organisation due to their larger cork and timber profitability. In recent years, there have also been funds promoting associations. But Lleida and Tarragona provinces lag behind in this regard. The analysed incentives bring clear associational benefits, and therefore their effects may encourage the spontaneous emergence of new associations in areas with lower forestry dynamism, such as the southern, more land-fragmented and less productive municipalities of Catalonia. This has been the case with two of the beneficiary FOAs. However, if there is no proactivity in those areas, we recommend external support to stimulate and catalyse new associations or groupings. In these contexts, organising owners into collective actions requires greater comparative effort (and consequent transaction costs [32]), which places them at a disadvantage when participating in the incentives. State support for changing such a payoff structure would be needed to boost collective action [5]. Otherwise, a perverse effect may occur, whereby these more marginalised forestry areas remain further relegated while they show a significant wildfire risk, as observed in less active yet riskier Portuguese ZIFs [32]. Forest owners' associations constitute a fundamental pillar in this process, but their structural fragility limits their long-term permanence. These associations rely heavily on the roles of the facilitator and the technician, who are key to their functioning. However, the complexity of administrative procedures and the high personal cost borne by facilitators can discourage engagement in such a key role.

Our results also show policymakers that by covering (at least partly) the coordination costs, spatial coordination is facilitated, given that the efforts towards catalysing landowners are at least partially rewarded. Moreover, by including zoning in the prioritisation criteria, effectiveness increases compared to traditional fuel reduction subsidies (see [10]). Nevertheless, economic incentives may be further targeted by explicitly requiring spatial adjacency, such as agglomeration bonuses [33]. Future research should explore this mechanism, which remains novel in the forestry domain.

A future challenge is the maintenance of low fuel levels in the treated areas. The financed fuel management is expected to be effective for bush removal during approximately five years, and for thinning about 10–15 years, with some (drier) areas lasting longer, depending on the rate of vegetation growth. The temporal commitment of the mechanisms is aligned with the thinning operations and, institutionally, with the duration of approved forest management plans; however, for effective influence in fire spread, it may require more frequent interventions in the understorey layer. Controlled grazing emerges as a potential economic activity to maintain the shrub and herbaceous layer; however, to be effective, it also requires payments to reward the additional efforts devoted by shepherds (see [34]). Moreover, if, thanks to one or two rounds of externally financed thinning, the forest becomes capitalised enough, with its landowners more engaged in active forestry, the ideal longer-term scenario would be that the timber itself would generate sufficient return to justify the investment in fuel treatments. Therefore, to ensure a reduced fuel landscape over time, there should be an expectation that these economic incentives will hold at sufficient budget levels beyond the previously mentioned permanence period. Subsidies will depend on political will regarding the topic, whereas climate credits rely on the continued will of funding companies. After the positive reception of the subsidies and

the catastrophic wildfires in 2025 in other Spanish regions, these fuel management subsidies appear likely to continue, at least in the short term. Regarding climate credits, the findings show that at least one quarter of the engaged companies had a multi-annual payment commitment. Yet, credit demand in the carbon voluntary market may become volatile if the business environment no longer values offsetting negative impacts from companies (e.g., [35]), such as CO_2 emissions, water footprint, or related biodiversity losses.

Future research could assess landowners' psychological effects of joint participation, with regard to levels of perceived security, satisfaction in relation to motivation objectives, perceived benefits of cooperation, sense of community belonging, or willingness to continue cooperation for other purposes. Further research using a larger temporal series of the analysed mechanisms could show whether the funds tend to complete the strategic zones (and thus become more effective) or instead spread across the territory, reaching new strategic zones without completing them. However, if economic incentives do not suffice to persuade landowners to become active and manage their forests to reduce a shared risk, alternatives need to be considered. These instruments do not occur in an institutional vacuum [5,35]; therefore, complementary instruments may be required to achieve the targeted cross-boundary fuel management coordination. In this regard, pushing for renting or selling, which tends to be preferred by less active Portuguese forest owners [35], or command-and-control policy mechanisms (e.g., land consolidation [16], or the South African liability for wildfire spread) may end up being applied, justified by the public benefit.

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Abbreviations

The following abbreviations are used in this manuscript:

FOA	Forest Owner Association
IOF	Forest Management Instrument
ORGEST	Guidance of Sustainable Forest Management
PROMACC	Forest Projects for Climate Change Mitigation and Adaptation
VAT	Value Added Tax
ZIF	Forest Intervention Zones

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